AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 23 · TEACHER GUIDE

Block to Python Bridge

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Teach equivalent procedures and different evidence records.

Model initialization, indentation and display; preserve actual errors and assess a fresh sum trace.

Keep one habit: Ask which line changes state, which line displays it and whether the result was predicted, traced or executed.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Translate a short paper calculation into a Python script

This is a first encounter with saved Python scripts, not a compressed implementation of the Grade 6 adviser. Teach **assignment**, multiplication and print only as needed; values and types follow in Week 2.

Use a fresh run to avoid hidden interactive state. Reference: Python tutorial, An Informal Introduction to Python, https://docs.python.org/3/tutorial/introduction.html (**assignment**, arithmetic and print).

By the end, I can explain block to python bridge using a traced example and a stated limitation.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Case exposure check: this source example appears in ai-2/1, ai-3/23, python-bridge/1. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.
- Teacher-prepared Python 3 editor with a new saved .py file and a visible **output** area; test Run before class
- Paper counters or equal-group sketches and a blank expected/observed record
- The printed script and workbook; no logins or personal data required

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Requests Responses and JSON. Give an example and a mistake that the idea helps you avoid.

Look for: Parse JSON into Python records

Reference example, not the only acceptable wording: JSON null becomes Python None and Python True serialises as JSON true. Parsing is followed by top-level, field, type and range validation; successful parsing does not prove a source observation true.

Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of conversation state and retention

Reference example, not the only acceptable wording: Teacher explanation: Conversation state stores facts or prior interactions for later use.

It needs scope, provenance, correction and retention rules rather than indefinite accumulation. This classroom assistant keeps an approved room ID only for the current session.

A later correction replaces the active value but retains an audit event until session end. Do not persist private text merely because it appeared in conversation.

Without opening the old lesson, explain one key idea from Model Memory vs Retrieval. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of retrieval and indexing

Reference example, not the only acceptable wording: Teacher explanation: Retrieval returns stored records selected for a query.

An index is a searchable representation of a versioned corpus; generated wording is a later operation. Assign stable document IDs, versions and permitted access before indexing. Retrieve only authorised records.

Preserve source references and define behaviour when a document is missing, outdated or access-restricted.

Record four and six, multiply them to get 24, then display 24. The order must make the required values available before use.

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Retrieval answer guidance

Requests Responses and JSON

Conversation State and Memory

Model Memory vs Retrieval

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Write Python that totals [3,5,7], then trace each total. Explain how indenting 'total=0' inside the loop changes the result.

Concept to explain

Variables, **condition**s, **list**s, loops, and **function**s carry the same logic across representations.

Term	Meaning
variable	A named reference to a value that can be used or changed in a program.
list	An ordered collection of items.
function	A named unit of code that performs a task, optionally receiving arguments and returning a value.
condition	A statement evaluated as true or false to help select an action.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach equivalent actions before syntax details

> Separate packing multiplication, qualifying counts and independent summation.

Set the short-script environment

In Grade 6 you traced rules on paper. Now you will write a small Python **program** and check what it actually does. Begin with a counter-packing calculation, not the whole activity adviser.

The requirement is to find the total counters in equal packs and display that total. For this first **program**, the supplied quantities are whole, nonnegative numbers; it does not yet validate typed **input**.

Use a teacher-prepared Python 3 editor that can run a saved .py file. Open a new file called counters.py, type the code exactly as shown in the next section, save it, and use the editor's Run command.

Read the **output** area separately from the source editor. No account, AI tool, package installation or personal data is needed. If a device is unavailable, trace on paper and label the result predicted rather than executed.

Read the script from top to bottom. The equals sign assigns the value on its right to the name on its left. The star means multiplication. print(total) displays the value of total; it does not display the letters in the name.

Type ordinary straight characters and keep the parentheses. The Python prompt symbols >>> are not part of this saved script.

Model the packing prediction record

Write 24 as the expected **output** before running. Then copy the actual **output** into a separate column. A matching result supports this case, not every possible **input**.

A saved file may contain an old edit: save again and rerun the whole script. Do not type the expected number into the **output** record without running.

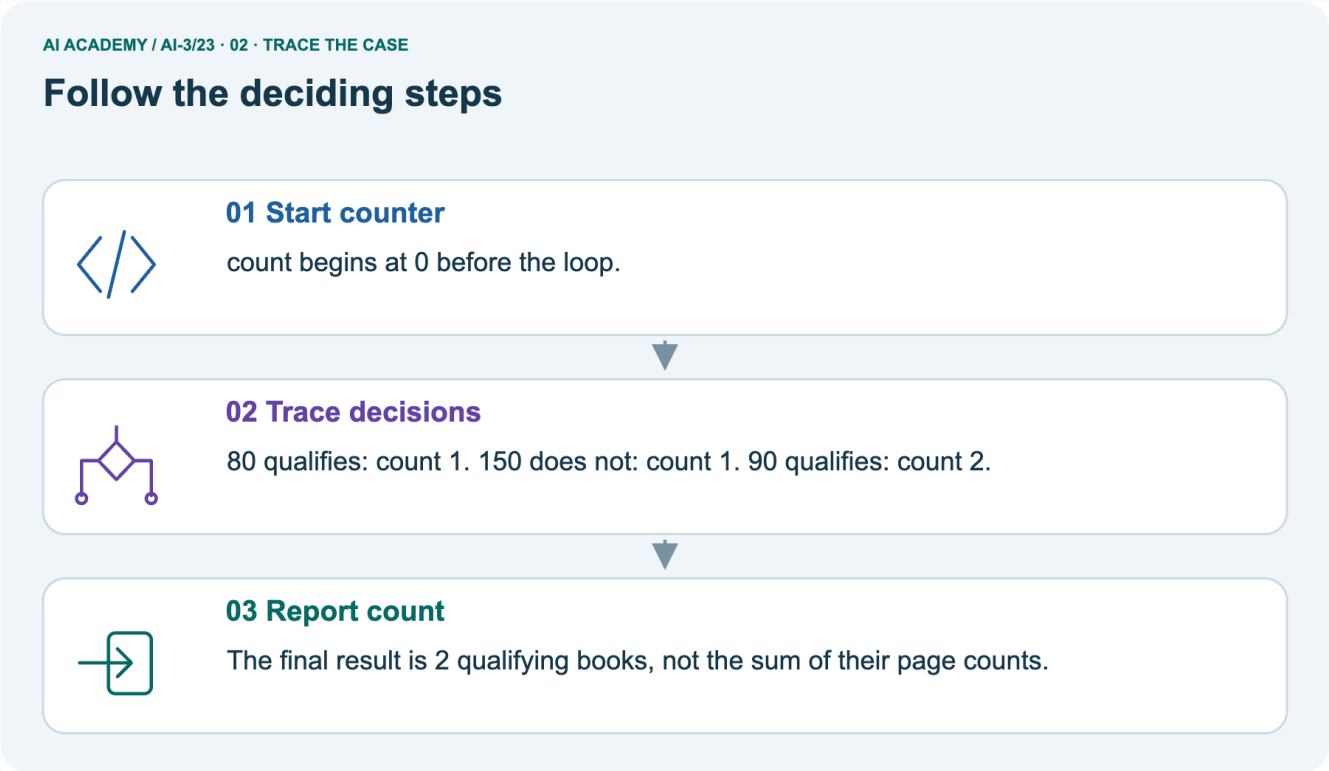
Read the book-count case

Translate a "count qualifying books" paper procedure into a Python-style trace.

Page counts are [80, 150, 90]. Start count=0; for each pages value, add 1 only when pages<100.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Trace the same procedure in picture and code



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Start counter	count begins at 0 before the loop.
2. Trace decisions	80 qualifies: count 1. 150 does not: count 1. 90 qualifies: count 2.
3. Report count	The final result is 2 qualifying books, not the sum of their page counts.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. The final result is 2 qualifying books, not the sum of their page counts. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Block action	Python and effect
Set counter once	count = 0 before the loop
Repeat for each book	for pages in page_counts:
Test the strict rule	if pages < 100:
Increase qualifying count	count = count + 1, inside the condition
Finish the function	return count after the loop; print the returned result

```
def count_short_books(page_counts):
    # Supplied input: a list of nonnegative integer page counts.
```

```
    count = 0
    for pages in page_counts:
        if pages < 100:
            count = count + 1
    return count

print(count_short_books([80, 150, 90]))
print(count_short_books([80, 150, 100]))
print(count_short_books([]))
```

Predict the strict boundary

Change: Replace the last page count 90 with 100.

Prediction: The final count becomes 1.

Reason: The comparison is strictly less than 100, so the boundary value does not qualify.

State the deterministic limitation

This deterministic loop counts a stated **condition**; it neither trains a model nor measures whether the books suit a reader.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied **condition** and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Additional practice · Different sum values

A block **program** sets total=0 and adds each value in [2,4,6].

Translate and trace.

Python uses total=0, then an indented total += value inside for value in [2,4,6]. Totals are 2,6,12.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Block: for each passage, check relevance.

2. Python: for passage in passages: ...

3. **Condition** keeps passages with score ≥ threshold.

4. A **function** packages the selection logic and can be tested with normal and edge **list**s.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Exact explanation to model

Explain the added worked example

Key terms

Term	Meaning
Algorithm	An ordered procedure for solving a stated problem
Program	Instructions written in a language a computer can execute
Interpreter	Software that executes Python instructions
Assignment	Binding a name to the value of an expression
Output	Information a program displays or returns

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

After line	Names and values	Displayed output
packs = 4	packs is 4	Nothing yet
per_pack = 6	packs is 4; per_pack is 6	Nothing yet
total = packs * per_pack	total is 24	Nothing yet
print(total)	Values are unchanged	24

> **Why is there no displayed output after the third line? What changes if print(total) is placed first in a fresh run, before any assignment? Explain using which names exist.**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Block to Python Bridge

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the explanation, then remove help

> Collect each learner's reasoning before showing feedback.

Main completion prompt

Completion task: Make records for 4 packs of 6, then 5 packs of 6. Predict before running. Record actual **output**, any error and what changed. If working on paper, label both results as traces.

Expected reasoning and fading help

Teacher check: Expected 24 then 30. Accept actual errors with an accurate account; do not replace **observation**s. Record the changed pack count, saved file and run or trace status.

Two matching cases do not show all **input**s are handled.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for From a paper **algorithm** to Python. Proposed explanation: "An equals sign displays the answer". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: **Assignment** binds a name; print produces the displayed **output**.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Trace the changed **list** and identify where the result differs. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

80 increments to 1; 150 leaves 1; 100 leaves 1. The last comparison changed from true to false. Changed-case check: The final count becomes 1. The comparison is strictly less than 100, so the boundary value does not qualify.

Changed-case reasoning: The final count becomes 1. The comparison is strictly less than 100, so the boundary value does not qualify.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: A block **program** sets total=0 and adds each value in [2,4,6].

Write the question to explain: Translate and trace.

Reveal after **prediction**s: Python uses total=0, then an indented total += value inside for value in [2,4,6]. Totals are 2,6,12.

Use the earlier changed question as optional later practice. Reserve Trace the new sum/reset procedure independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace **condition**s for decisions, and identify supporting sources for claims.

Model these guided questions

The third line assigns total but does not call print, so nothing is displayed. A first-line print(total) raises NameError in a fresh run because total has not been assigned.

Exit: 11 is wrong for five groups of six; addition is a likely cause, and multiplication should give 30. Retain the old result and check the repaired file with both pack counts. Record whether the work was executed or traced.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve equal-group multiplication and collect the prerequisite explanation.
7 to 20 minutes	Model source versus output, save counters.py and translate each paper step with the diagram.
20 to 33 minutes	Complete W1 and predict the trace before running the four-line program.
33 to 45 minutes	Start W2 and distinguish observation from the expected answer; record device access.
Session 2 start to 8 minutes	Rerun with five packs and compare the two records.
8 to 20 minutes	Use W3 and the guided question to diagnose a first-line print in a fresh run.

Stage	Teacher action
20 to 35 minutes	Collect W4 independently before hints; observe whether each pupil can explain zero trays.
35 to 45 minutes	Complete W5 and the exit. Save source and records; identify pupils needing editor support.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

Packing outputs 24 and 30 are predictions until run; paper traces must be labelled. Book counts 2, 1, 0 for supplied normal/boundary/empty lists. Record D has pages 80 and qualifies; E lacks pages, so no count can be inferred from an invented zero. Independent sum: totals 2, 6, 15; resetting before each addition gives 2, 4, 9 and final 9. Deterministic code does not train a model.

Run and record

1. Predict → Test → Record → Explain

2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.

3. Use the complete five-action book-count mapping and supplied cases A–E. Run or trace numeric **list**s, including empty and boundary cases. For record cards, explain the missing-pages guard separately; do not claim the numeric-**list** function parses records or validates arbitrary input.

4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

An equals sign displays the answer: **Assignment** binds a name; print produces the displayed **output**.

A successful run means the requirement was met: Compare the **output** with the paper rule and expected result.

Python guesses the intended order: A name must be assigned before this script reads it.

Typing a **program** trains an AI model: These statements apply an explicit multiplication rule.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Trace the new sum/reset procedure independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected sum/reset trace

> Reveal the selected key after the first attempt; packing transfer remains additional practice.

Selected independent question and key

Selected case: Set total=0, then for values2,4,9 perform total=total+value. Trace the result. What changes if total=0 moves inside the loop before that addition?

Running totals are2,6,15. Resetting inside the loop produces2,4,9, leaving9. Initialization location changes the written procedure; neither version trains a model.

Judge trace and scope

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

▼ Packing workbook guidance

W1 Connect the paper steps to Python

Write the complete four-line script for 4 packs of 6. Beside each line explain what it does. Which line displays information?

packs = 4; per_pack = 6; total = packs * per_pack; print(total), each on its own line. The first two bind quantities, the third computes 24 and the fourth displays 24.

Assignment alone produces no displayed **output** in this script.

W2 Predict then observe

Make records for 4 packs of 6, then 5 packs of 6. Predict before running. Record actual **output**, any error and what changed. If working on paper, label both results as traces.

Expected 24 then 30. Accept actual errors with an accurate account; do not replace **observation**s. Record the changed pack count, saved file and run or trace status. Two matching cases do not show all **input**s are handled.

W3 Explain an order error

Move print(total) to the first line of a fresh script. Predict what happens and why. Put the lines back into an order that meets the requirement.

The first line tries to read total before it exists, causing NameError in a fresh run. Assign packs and per_pack, calculate total, then print it.

Earlier values lingering in an interactive session can hide the problem; run the complete saved script afresh.

W4 Transfer the procedure independently

A fictional shelf has 3 trays of 7 tiles each. Write a four-line Python script using trays, per_tray and tiles. Predict its **output**. Then change only trays to 0 and predict again. Explain why adding the two quantities is wrong.

trays = 3; per_tray = 7; tiles = trays * per_tray; print(tiles), on separate lines. **Output**s 21 and then 0. Zero trays contain zero tiles. Addition would give 10 then 7, not the number in equal groups.

W5 Keep an honest conclusion

Explain the difference between a successful run and a correct calculation. State one limit of this **program** and what you would keep so another person can check your work.

A successful run means the execution finished without a reported error; correctness requires matching the stated total-count rule. The script assumes supplied whole nonnegative quantities and does not validate arbitrary **input**.

Keep source, quantities, expected and actual **output** and whether it was run or traced.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Python uses total=0, then an indented total += value inside for value in [2,4,6]. Totals are 2,6,12.

Additional transfer practice question

Trace this Python: score = 2; then if score >= 2: score = score + 3. What is the final value, and what changes if the **condition** becomes score > 2?

Reasoned practice answer

The original **condition** is true, so score becomes 5. With score > 2, the **condition** is false and score stays 2.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Greater than and greater than or equal to behave identically at the boundary.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Use total=0, then for value in [3,5,7]: with indented total += value. Totals are 3,8,15. Resetting to zero inside each iteration leaves 7, the last value. Trace the empty **list** too: the correctly initialised result is 0.

Explanations for the additional workbook examples

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Implement small transparent **function**s for selection, validation, and citation formatting.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support while preserving the reasoning goal

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Offer four separate line cards, a source/**output** label and a multiplication sketch. Allow a paper trace when devices are unavailable, with execution practice still pending.

Do not supply the independent transfer numbers' answers. Extension: explain why a plus-sign version can accidentally agree with multiplication for two packs of two, and why one passing example is weak **evidence**.

Selected independent task: running totals 2, 6, 15. Resetting total to 0 inside the loop before addition gives 2, 4, 9 and final 9. Require a trace of both placements. Fresh retry: sum 3, 5, 8 gives 3, 8, 16; resetting each time gives 3, 5, 8 and final 8. Keep paper reasoning and actual execution separate; neither procedure trains a model. Earlier packing tasks remain additional practice.

Plan the delayed check

Suggested return points in this level: ai-3/24, ai-3/26, ai-3/29. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 24: Prototype a Grounded Assistant. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 24: Prototype a Grounded Assistant. Carry forward small readable procedures, explicit inputs and outputs, and separate predictions from actual test evidence.

▼ **Earlier wording, alternatives and picture transcripts**

Connect and extend learning

Verified Study Buddy

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Translate a short paper calculation into a Python script

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

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Practice and assessment reminders

Before class, solve the supported practice: Make records for 4 packs of 6, then 5 packs of 6. Predict before running. Record actual **output**, any error and what changed. If working on paper, label both results as traces.

Reasoned practice answer: Expected 24 then 30. Accept actual errors with an accurate account; do not replace **observation**s. Record the changed pack count, saved file and run or trace status.

Two matching cases do not show all **input**s are handled.

Check every learner before discussion: Use the worked case for From a paper **algorithm** to Python. Proposed explanation: “An equals sign displays the answer”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: **Assignment** binds a name; print produces the displayed **output**.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Replace the last page count 90 with 100. Require a **prediction** and a reason before comparing. Expected result: The final count becomes 1.

Reason: The comparison is strictly less than 100, so the boundary value does not qualify.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

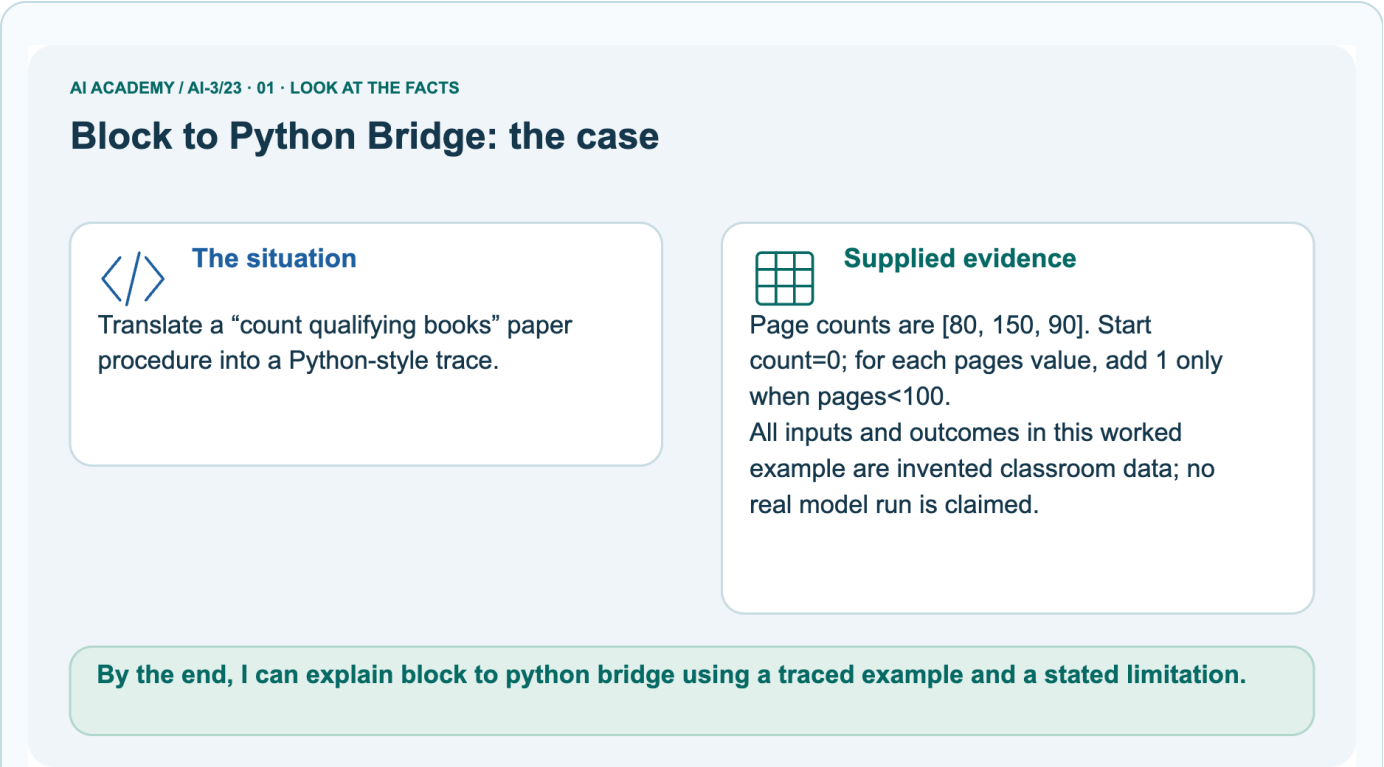
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Block to Python Bridge: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

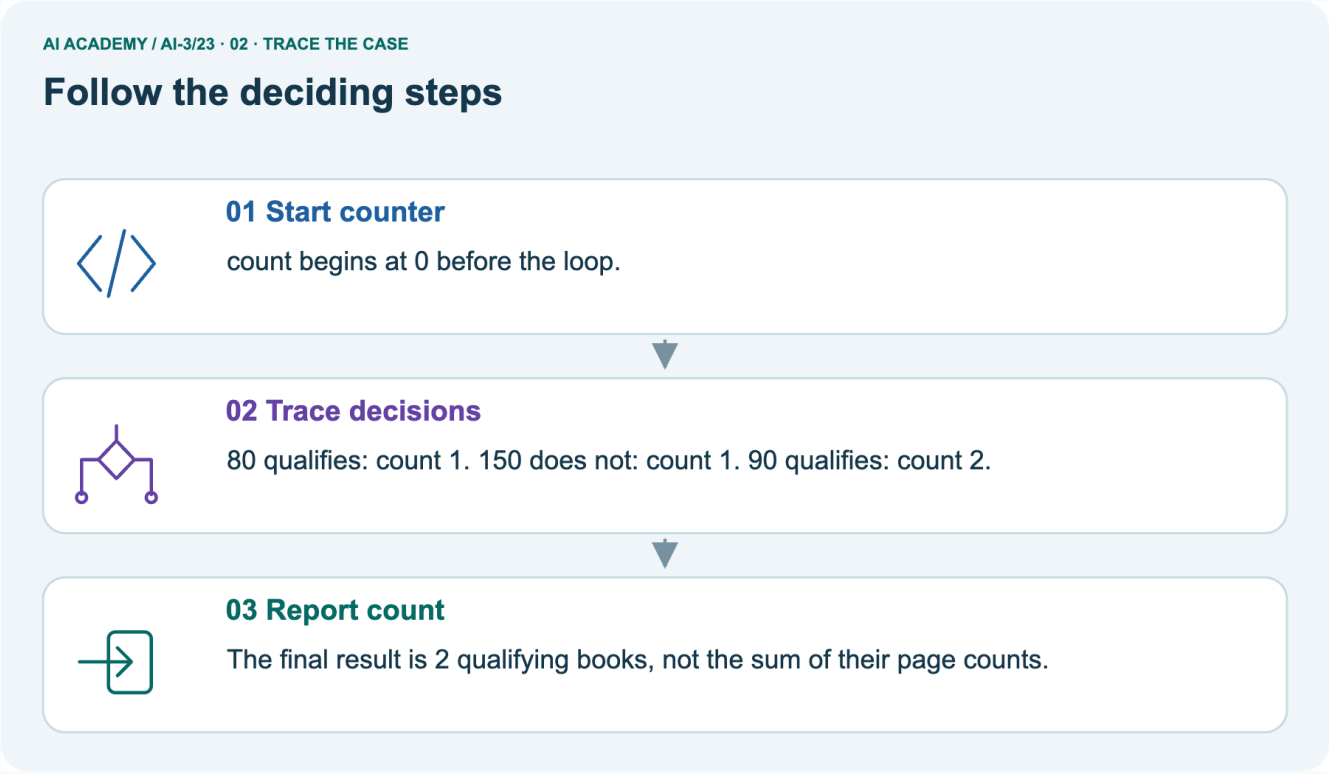
Translate a “count qualifying books” paper procedure into a Python-style trace. Supplied facts: Page counts are [80, 150, 90]. Start count=0; for each pages value, add 1 only when pages<100.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



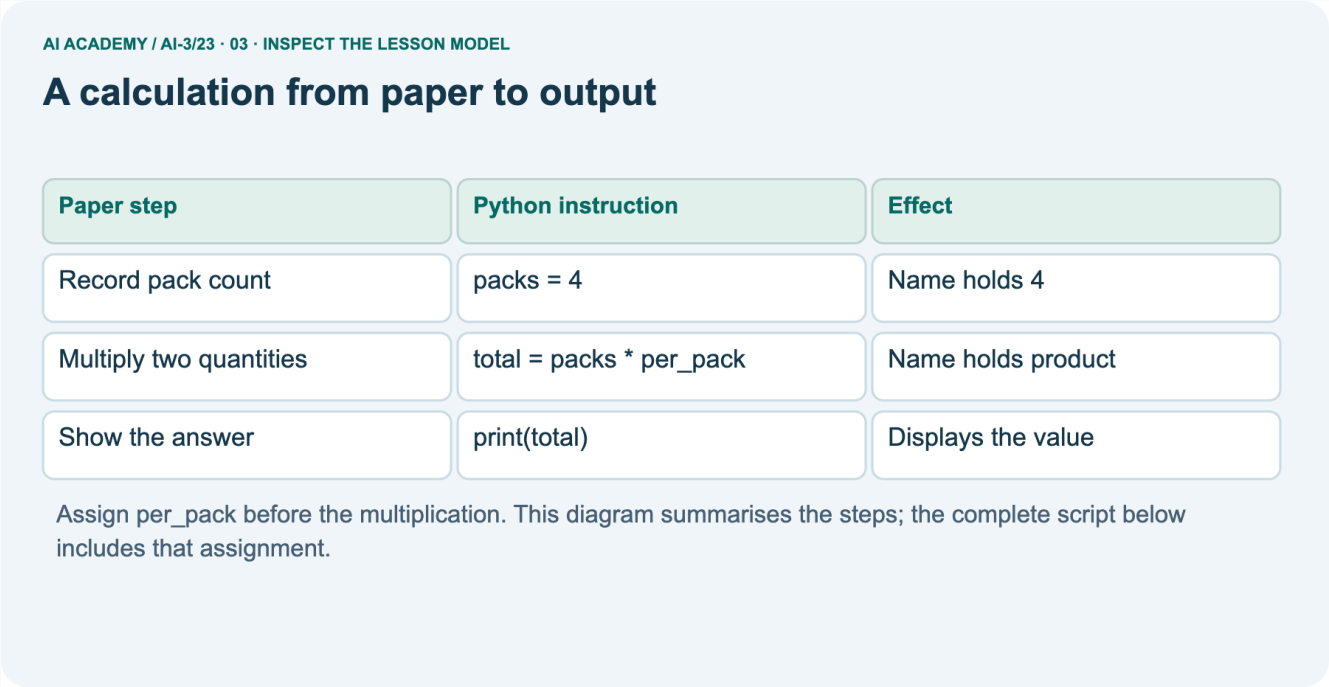
Follow the labelled picture with the matching explanation. Open the vector diagram

Start counter: count begins at 0 before the loop. → Trace decisions: 80 qualifies: count 1. 150 does not: count 1. 90 qualifies: count 2. → Report count: The final result is 2 qualifying books, not the sum of their page counts.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. A calculation from paper to output



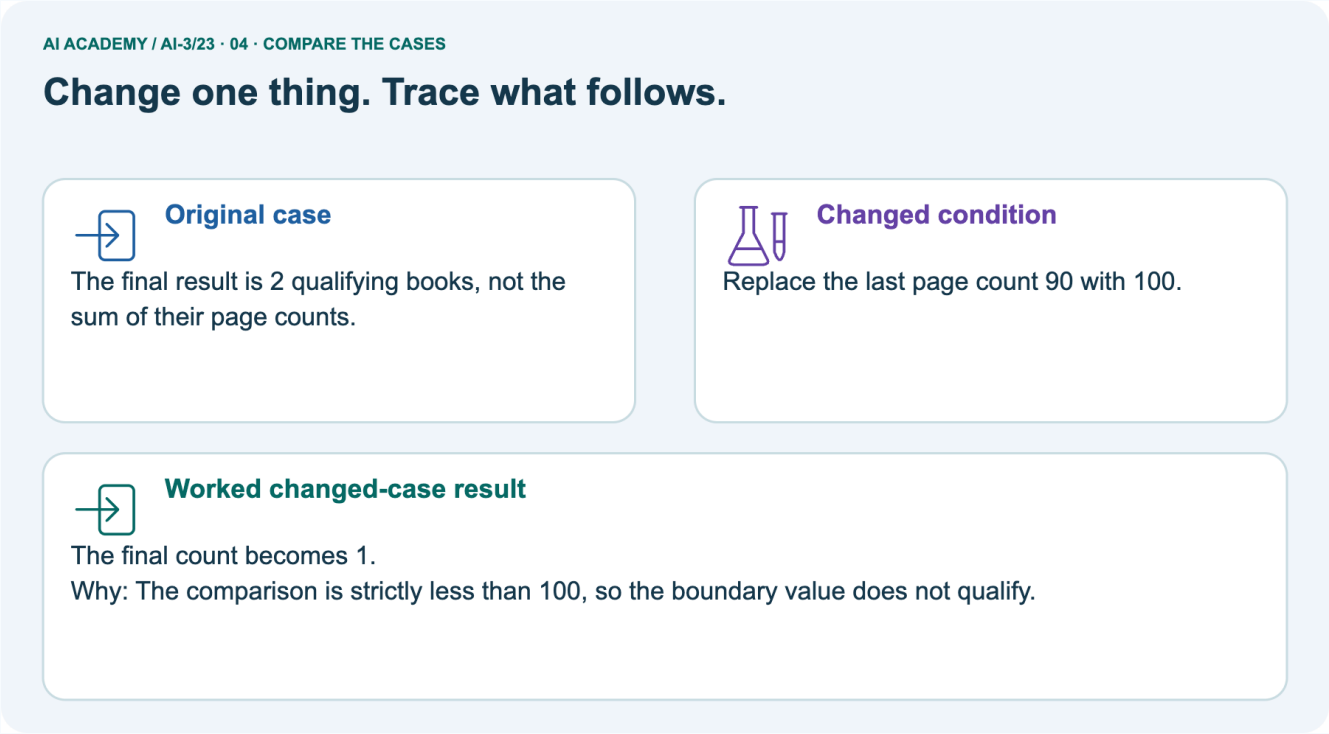
Follow the labelled picture with the matching explanation. Open the vector diagram

A calculation from paper to **output**. Record pack count | packs = 4 | Name holds 4. Multiply two quantities | total = packs * per_pack | Name holds product. Show the answer | print(total) | Displays the value

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The final result is 2 qualifying books, not the sum of their page counts. Changed **condition**: Replace the last page count 90 with 100. Expected result: The final count becomes 1.

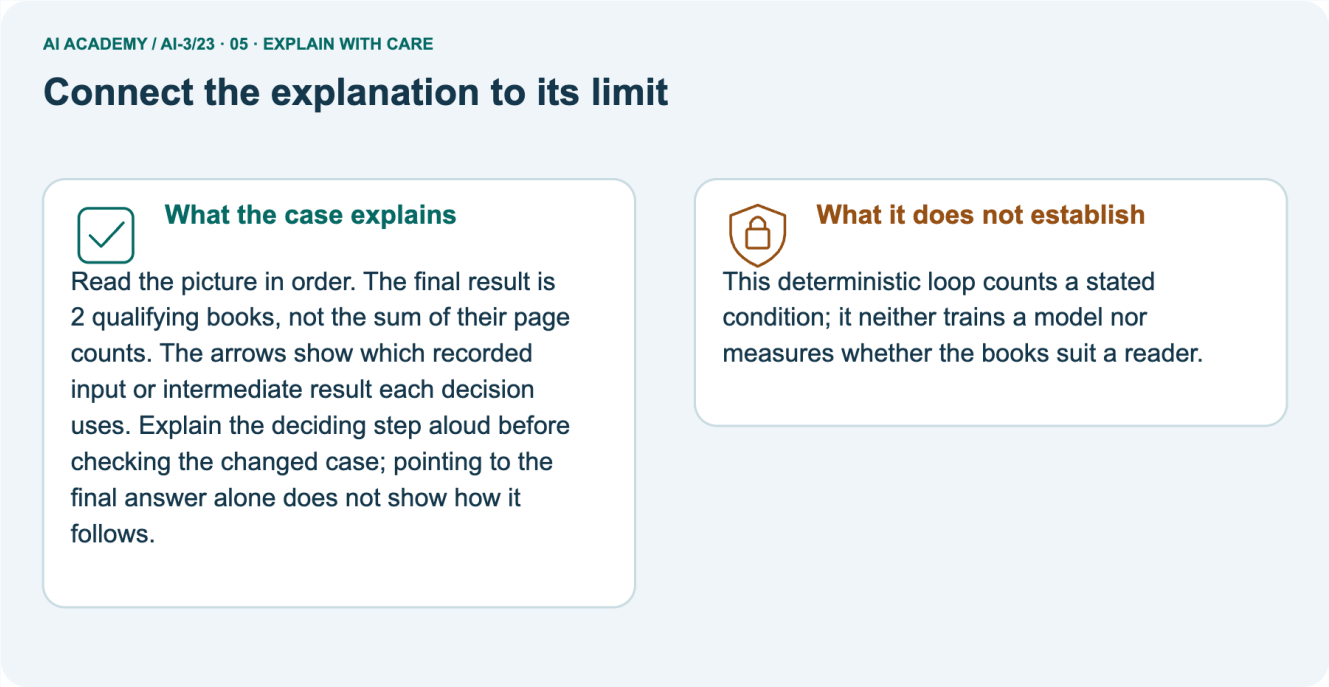
Reason: The comparison is strictly less than 100, so the boundary value does not qualify.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The comparison is strictly less than 100, so the boundary value does not qualify.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. The final result is 2 qualifying books, not the sum of their page counts. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

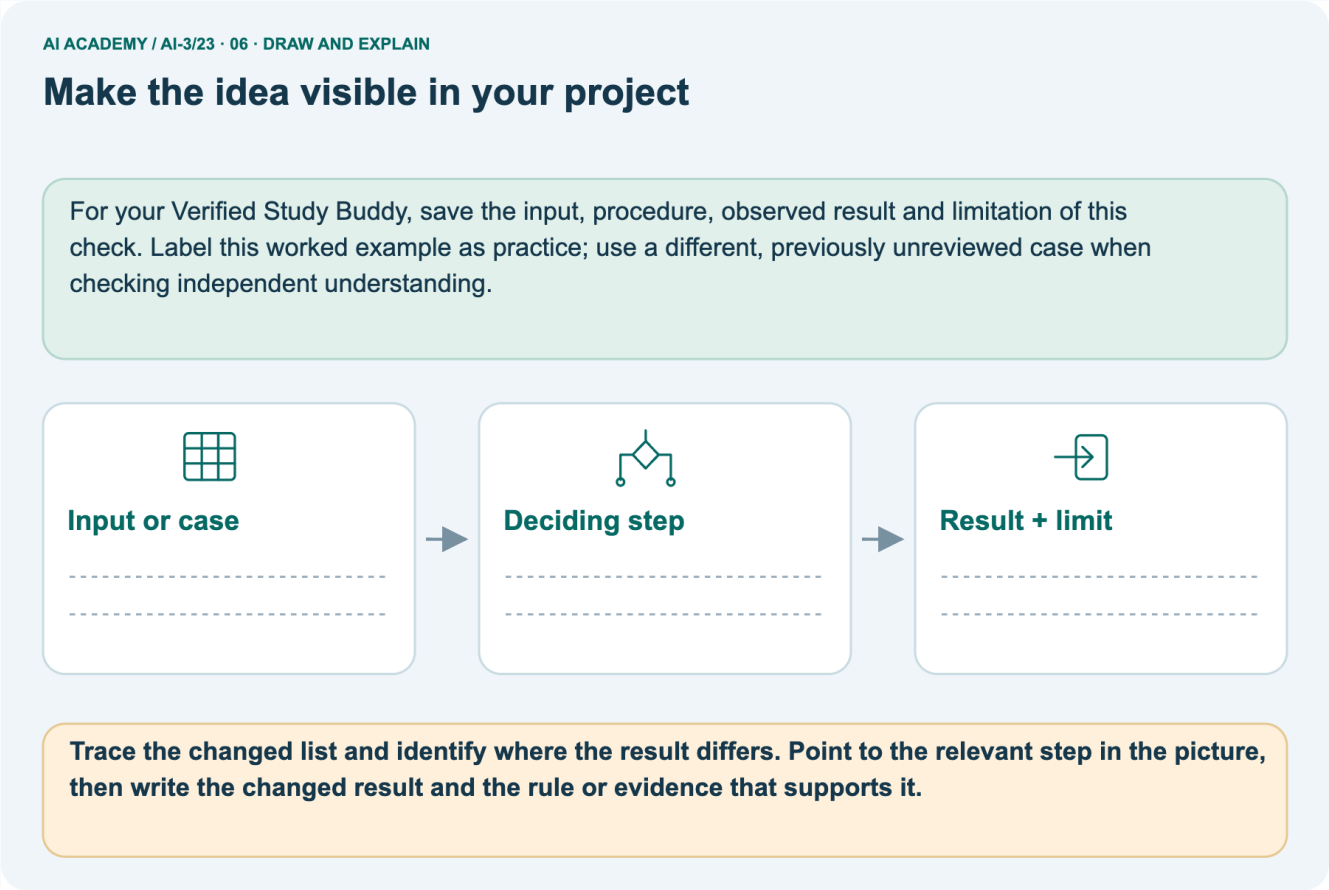
Limit: This deterministic loop counts a stated **condition**; it neither trains a model nor measures whether the books suit a reader.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This deterministic loop counts a stated **condition**; it neither trains a model nor measures whether the books suit a reader.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Trace the changed **list** and identify where the result differs. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: 80 increments to 1; 150 leaves 1; 100 leaves 1. The last comparison changed from true to false. Changed-case check: The final count becomes 1.

The comparison is strictly less than 100, so the boundary value does not qualify.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Block to Python Bridge: the case

02. Follow the deciding steps

03. A calculation from paper to output

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.